$$=\sqrt{\left[2\gamma_m(2n_T+1)+\frac{\gamma_m^2+\frac{1}{3}\left[\frac{2\pi}{\tau}\right]^2}{\mathcal{K}}+\mathcal{K}\right]\frac{1}{\tau}} \tag{C2}$$

Here we assume that angle $\psi_f=0$ in (3.10), and $\mathcal{K}$ is a constant. We consider the case of short time τ of force action:

$$\gamma_m\tau \ll 1 \tag{C3}$$

(opposite case of large τ is not interesting, thermal limit restricts sensitivity).

Taking minimum over pump $\mathcal{K}$ with account of (C3), we obtain minimum force F_{s0} to be detected:

$$\frac{F_{s0}^2}{4\hbar m\omega_m} \geq 2\gamma_m(2n_T+1)\,\frac{1}{\tau}+\frac{2}{\sqrt{3}}\frac{2\pi}{\tau^2} \tag{C4}$$

Here first term describes thermal limit whereas second one — SQL:

$$F_{s0}^{SQL}=\frac{4}{\tau}\sqrt{\frac{\pi\hbar m\omega_m}{\sqrt{3}}} \tag{C5}$$

This formula is valid with accuracy of constant multiplier about 1 due to approximation of (C1).

We can take (3.25) instead of (3.24), then we obtain condition (C3)

$$\frac{\tilde{F}_{s0}^2}{4\hbar m\omega_m} \geq 2\gamma_m(2n_T+1)\,\frac{1}{\tau}+\frac{4\pi}{\tau^2} \tag{C6}$$

Last terms in (C4) and in (C6) differ only by multiplier about 1. So in frequency domain we can use spectral density (3.25) for SQL characterization.

Appendix D: Input-output relations (conventional squeezing)

Here we present detail formulas for derivation of output quadratures (4.3) for case of standard squeezing.

We obtain set for intracavity amplitudes in frequency domain:

$$(\gamma-i\Omega)c_+(\Omega)=-\eta C_0 d(\Omega)-\upsilon c_+^\dagger(-\Omega)+ \tag{D1a}$$
$$+\sqrt{2\gamma_0}\,a_+(\Omega)+\sqrt{2\gamma_e}e_+(\Omega),$$
$$(\gamma-i\Omega)c_-(\Omega)=\eta C_0 d^\dagger(-\Omega)-\upsilon c_-^\dagger(-\Omega)+ \tag{D1b}$$
$$+\sqrt{2\gamma_0}\,a_-(\Omega)+\sqrt{2\gamma_e}\,e_-(\Omega),$$
$$(\gamma_m-i\Omega)d(\Omega)=\eta C_0\left[c_-^\dagger(-\Omega)+c_+(\Omega)\right]+ \tag{D1c}$$
$$+\sqrt{2\gamma_m}\,\hat{q}(\Omega)+f_s(\Omega).$$

These equations differ from set (B5) only by terms $\sim\upsilon$.

Then using (D1) we obtain for sum and difference of quadratures

$$(\gamma-i\Omega)c_{+a}+\upsilon c_{+a}+\eta C_0 d_a= \tag{D2a}$$
$$=\sqrt{2\gamma_0}a_{+a}+\sqrt{2\gamma_e}e_{+a},$$
$$(\gamma-i\Omega)c_{+\phi}-\upsilon c_{+\phi}+\eta C_0 d_\phi= \tag{D2b}$$
$$+\sqrt{2\gamma_0}a_{+\phi}+\sqrt{2\gamma_e}e_{+\phi},$$
$$(\gamma-i\Omega)c_{-a}+\upsilon c_{-a}-\eta C_0 d_a= \tag{D2c}$$
$$=\sqrt{2\gamma_0}a_{-a}+\sqrt{2\gamma_e}e_{-a},$$
$$(\gamma-i\Omega)c_{-\phi}-\upsilon c_{-\phi}+\eta C_0 d_\phi= \tag{D2d}$$
$$=\sqrt{2\gamma_0}a_{-\phi}+\sqrt{2\gamma_e}e_{-\phi},$$
$$(\gamma_m-i\Omega)d_a-\eta C_0\Big(c_{+a}+c_{-a}\Big)= \tag{D2e}$$
$$=\sqrt{2\gamma_m}q_a+f_{s\,a},$$
$$(\gamma_m-i\Omega)d_\phi-\eta C_0\Big(c_{+\phi}-c_{-\phi}\Big)= \tag{D2f}$$
$$=\sqrt{2\gamma_m}q_\phi+f_{s\,\phi}.$$

This set differs from (B6) only by terms $\sim\upsilon$.

Again, sum $(c_{+a}+c_{-a})$ does not contain information on the mechanical motion (term proportional to $\sim d_a$ is absent), but produces the back action term in (D2e). At the same time difference $(c_{+\phi}-c_{-\phi})$ does not contain information on mechanical motion (term $\sim d_\phi$), but is responsible on back action in (D2f). So it should be useful to introduce sum and difference of the quadratures (3.14) and rewrite (D2) in the new notations

$$(\gamma+\upsilon-i\Omega)g_{a+}=\sqrt{2\gamma_0}\alpha_{a+}+\sqrt{2\gamma_e}\epsilon_{a+}, \tag{D3a}$$
$$(\gamma+\upsilon-i\Omega)g_{a-}+\sqrt{2}\eta C_0 d_a=\sqrt{2\gamma_0}\alpha_{a-}+ \tag{D3b}$$
$$+\sqrt{2\gamma_e}\epsilon_{a-},$$
$$(\gamma_m-i\Omega)d_a-\sqrt{2}\eta C_0 g_{a+}=\sqrt{2\gamma_m}q_a+f_{s\,a}, \tag{D3c}$$
$$(\gamma-\upsilon-i\Omega)g_{\phi+}+\sqrt{2}\eta C_0 d_\phi=\sqrt{2\gamma_0}\alpha_{\phi+}+ \tag{D3d}$$
$$+\sqrt{2\gamma_e}\epsilon_{\phi+},$$
$$(\gamma-\upsilon-i\Omega)g_{\phi-}=\sqrt{2\gamma_0}\alpha_{\phi-}+\sqrt{2\gamma_e}\epsilon_{\phi-}, \tag{D3e}$$
$$(\gamma_m-i\Omega)d_\phi-\sqrt{2}\eta C_0 g_{\phi-}=\sqrt{2\gamma_m}q_\phi+f_{s\,\phi}. \tag{D3f}$$

Again, the sets (D3a, D3b, D3c) for amplitude quadratures and (D3d, D3e, D3f) for phase quadratures are independent and can be separated.

As example, let consider the set for amplitude quadratures. The main difference, as compared with (B7), is that *both* g_{a+} and g_{a-} are squeezed by similar way, whereas in (B7) — by opposite way (g_{a+} is unsqueezed and g_{a-} is squeezed).

Using (3.3) and (3.14a) one can derive equation for β_{a-}

$$\beta_{a\pm}=\sqrt{2\gamma_0}\,\alpha_{a\pm}-\alpha_{a\pm},\quad \gamma\equiv\gamma_0+\gamma_e \tag{D4a}$$
$$\beta_{a+}=\frac{\gamma_0-\gamma_e-\upsilon+i\Omega}{\gamma_0+\gamma_e+\upsilon-i\Omega}\cdot\alpha_{a+}+\frac{2\sqrt{\gamma_0\gamma_e}\,\epsilon_{a+}}{\gamma+\upsilon-i\Omega}, \tag{D4b}$$
$$\beta_{a-}=\frac{\gamma_0-\gamma_e-\upsilon+i\Omega}{\gamma_0+\gamma_e+\upsilon-i\Omega}\cdot\alpha_{a-}+\frac{2\sqrt{\gamma_0\gamma_e}\,\epsilon_{a-}}{\gamma+\upsilon-i\Omega}- \tag{D4c}$$
$$-\frac{2\sqrt{\gamma}_0\,\eta C_0}{(\gamma+\upsilon-i\Omega)}\,d_a$$

Substituting (D3c) (with (D4b)) into (D4c), we finally rewrite equation for β_{a-}

$$\beta_{a-} = \frac{\gamma_0 - \gamma_e - \upsilon + i\Omega}{\gamma_0 + \gamma_e + \upsilon - i\Omega} \cdot \alpha_{a-} + \frac{2\sqrt{\gamma_0\gamma_e}\,\epsilon_{a-}}{\gamma + \upsilon - i\Omega} \tag{D5a}$$

$$- \frac{4\gamma_0\,\eta^2 C_0^2}{(\gamma + \upsilon - i\Omega)^2(\gamma_m - i\Omega)} \left(\alpha_{a+} + \sqrt{\frac{\gamma_e}{\gamma_0}}\,\epsilon_{a+}\right) - \frac{2\sqrt{\gamma}_0\,\eta C_0}{(\gamma + \upsilon - i\Omega)(\gamma_m - i\Omega)} \left(\sqrt{2\gamma_m} q_a + f_{s\,a}\right). \tag{D5b}$$

and finally obtain equations (4.3) for sum and difference amplitude quadratures $\beta_{a\pm}$.

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